Biochemistry Cumulative Examination
12/05/2014

In your structures drawing for this exam you may use Fischer, Haworth, a plain old skeletal structure, or whatever style you prefer, and switch back and forth as you wish, but in all your drawings the stereochemistry must be clear.

1. What sugar(s) make(s) up sucrose? 10 points

2. Draw the structure of glucose. 11 points. Is the common form D or L? 3 points. Which did you draw? 1 point.

3. Draw the structure of N-acetylglucosamine. 15 points

4. Where are some places one commonly finds N-acetylglucosamine? 15 points

5. What sugar is found in DNA? Name it. 3 points. Draw the structure. 12 points

6. What are the two simplest (smallest) sugars typically commonly found in cells (albeit usually phosphorylated)? Name and draw each in unphosphorylated form. 20 points

7. What is the name we use for the carbon in a sugar that often has two oxygens bonded to it? 10 points